

Joint Waveform Design and Beampattern Control for FDA-MIMO Dual-Function Radar-Communication Systems

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Abstract—This paper studies joint waveform design and transmit beampattern control for frequency diverse array (FDA)-multiple-input multiple-output (MIMO) dual-function radar-communication (DFRC) systems. By exploiting the time-range-angle coupling induced by frequency increments, a time-varying beampattern is shaped to enhance radar low-probability-of-intercept performance while embedding multi-user communication information. The design is formulated as a min-max beampattern optimization subject to radar SINR, constant-modulus waveform, and symbol-level communication constraints. To address the resulting non-convex problem, an alternating optimization algorithm is developed, where the radar receive beamformer admits a closed-form MVDR solution and the transmit waveform is obtained via a two-step SDR-based convex optimization approach. Simulation results demonstrate the effectiveness of the proposed method.

Index Terms—Dual-function radar-communication, frequency diversity array, multiple-input multiple-output, low probability of interception, waveform design.

I. INTRODUCTION

Dual-function radar-communication (DFRC) systems enable the integration of sensing and data transmission using shared spectral and hardware resources, thereby offering improved spectral efficiency, reduced system cost, and enhanced operational flexibility [1], [2]. However, these benefits come at the expense of increased design complexity, since reliable radar detection and low probability of intercept (LPI) performance must be jointly maintained under stringent multi-user communication and waveform constraints [3], [4]. In this context, frequency diverse array (FDA) multiple-input multiple-output (MIMO) systems introduce an additional design dimension, yielding transmit beampatterns that are inherently dependent on both angle and range [5]. This property makes FDA-MIMO particularly appealing for DFRC applications, as it provides enhanced spatial selectivity and improved interference suppression capability [6].

This work is supported in part by the National Natural Science Foundation of China under Grant 62271247, in part by the Natural Science Foundation of Jiangsu Province under Grant BK20240181, in part by Qing Lan Project of Jiangsu Province, and in part by Postgraduate Research and Practice Innovation Program of Jiangsu Province under Grant KYCX25_0592.

The existing FDA-MIMO waveform design approaches are formulated within snapshot-based or time-invariant beamforming optimization frameworks. By carefully designing the FDA-MIMO radar ambiguity function, the Doppler tolerance characteristics of the transmitted waveform can be applied to enhance the detection capability of moving targets [7]. The FDA-MIMO transmission waveforms and the received beamformer can be collaboratively designed to strengthen suppression capability against deceptive interference [8]. In addition, FDA-MIMO radar and MIMO communication systems have been investigated under spectrum coexistence paradigms, where the transmitted waveforms are treated as mutual interference to realize simultaneous sensing and data transmission [9]. However, these methods overlook the intrinsic fast-time variations induced by frequency offsets and waveform phase coupling in transmissions of the FDA signal, resulting in insufficient control over the temporal evolution of the transmit beampattern and potential degradation of the LPI performance.

To overcome these limitations, this paper proposes a joint waveform design and beampattern control method for FDA-MIMO DFRC systems. By explicitly accounting for the frequency-increment-induced time-range-angle coupling, the proposed approach enables flexible time-varying beampattern shaping aimed at suppressing instantaneous transmit power peaks and enhancing LPI performance. An alternating optimization algorithm is developed, where the radar receive beamformer admits a closed-form solution based on the minimum variance distortionless response (MVDR) criterion, and the transmit waveform is efficiently optimized through a two-step semidefinite relaxation (SDR)-based procedure. Numerical results demonstrate that the proposed framework achieves effective time-varying beampattern control while ensuring robust radar detection and stable communication performance.

II. SYSTEM MODEL

A. Transmit signal model

A FDA-MIMO-based DFRC transmitter featuring a one-dimensional uniform linear array (ULA) is considered, consisting of N_t transmitting antennas with a half-wavelength spacing d between each antenna. With f_0 being the reference

frequency, the transmission frequency of the m -th antenna is $f_m = f_0 + (m-1)\Delta f, m = 1, \dots, N_t$. Moreover, let $\mathbf{x}(t) = [x_1(t), \dots, x_{N_t}(t)]^T \in \mathbb{C}^{N_t}$ denotes the narrow-band transmit waveform with bandwidth B_x , where t represents the fast time. The transmit signal of the FDA-MIMO DFRC array can be expressed as

$$\mathbf{s}(t) = \mathbf{x}(t) \odot \mathbf{a}_f(\Delta f, t), \quad (1)$$

where $\mathbf{a}_f(\Delta f, t) = [1, \dots, e^{j2\pi(N_t-1)\Delta f t}]^T \in \mathbb{C}^{N_t}$ is the frequency increment vector, with f_0 being disregarded.

Accordingly, the instantaneous transmit beampattern at the far-field direction (R, θ) can be derived as

$$P_{\text{tran}}(R, \theta, \Delta f, t) = \mathbf{a}_t^H(\theta) \mathbf{s}\left(t - \frac{R}{c}\right) \mathbf{s}^H\left(t - \frac{R}{c}\right) \mathbf{a}_t(\theta), \quad (2)$$

where $\mathbf{a}_t \in \mathbb{C}^{N_t}$ represents the transmit steering vector.

To apply waveform design-based time-varying transmitted beampattern control, assuming the fast-time line consists of N timeslots, the n -th waveform is $\mathbf{x}[n] = [x_1[n], \dots, x_{N_t}[n]]^T \in \mathbb{C}^{N_t}$. Then, the transmit beampattern of the n -th timeslot can be expressed as

$$P_{\text{tran}}(R_t, \theta_t, \Delta f, \mathbf{x}[n]) = \mathbf{x}^H[n] \mathbf{A}_{t,n} \mathbf{x}[n], \quad (3)$$

where $\mathbf{A}_{t,n}(i, j) = e^{j\pi(j-i)\sin(\theta_t)} e^{j2\pi(i-j)\Delta f\left(nT_s - \frac{R_t}{c}\right)}$, and $T_s = \frac{T_p}{N}$ with T_p being the pulse duration.

B. Radar model

The FDA-MIMO DFRC system employs a colocated transmit-receive architecture, where the radar receive array consists of N_r antennas and is aligned with the FDA-MIMO transmit array. The radar scene comprises a desired target and multiple stationary clutter scatterers. The target is located at (R_t, θ_t) , while Q clutter scatterers are located at $(R_q, \theta_q), q = 1, \dots, Q$. The radar received data $\mathbf{R}_r \in \mathbb{C}^{N_r N}$ can be expressed as

$$\mathbf{R}_r = \gamma_t \mathbf{V}(\Delta f, R_t, \theta_t) \mathbf{X} + \sum_{q=1}^Q \gamma_q \mathbf{V}(\Delta f, R_q, \theta_q) \mathbf{X} + \mathbf{N}_r, \quad (4)$$

where

- $\mathbf{X} = [\mathbf{x}^T[1], \dots, \mathbf{x}^T[N]]^T \in \mathbb{C}^{N_t N}$,
- γ_t and γ_q are power coefficients of the target echo and clutter,
- $\mathbf{a}_r(\theta) \in \mathbb{C}^{N_r}$ is radar receive steering vector,
- $\mathbf{V}(\Delta f, R, \theta) = \mathbf{I}_N \otimes (\mathbf{a}_r(\theta) \mathbf{A}^H(\Delta f, R, \theta)) \in \mathbb{C}^{N_r N \times N_t N}$,
- $\mathbf{A}(\Delta f, R, \theta) = \mathbf{a}_t(\theta) \odot \mathbf{a}_f(\Delta f, -\frac{R}{c}) \in \mathbb{C}^{N_t}$,
- $\mathbf{N}_r \in \mathbb{C}^{N_r N}$ is the radar receive noise.

Then, the radar implements the digital beamforming to the received data of (4), which formulates the output signal-to-interference-plus-noise ratio (SINR) as

$$\text{SINR}(\mathbf{W}, \mathbf{X}) = \frac{\gamma_t^2 \mathbf{W}^H \mathbf{R}_{t,x} \mathbf{W}}{\mathbf{W}^H \mathbf{R}_{n,x} \mathbf{W}} = \frac{\gamma_t^2 \mathbf{X}^H \mathbf{R}_{t,u} \mathbf{X}}{\mathbf{X}^H \mathbf{R}_{n,u} \mathbf{X}}, \quad (5)$$

where

- $\mathbf{W} = [\mathbf{W}^T[1], \dots, \mathbf{W}^T[N]]^T \in \mathbb{C}^{N_r N}$, $\mathbf{W}[n] \in \mathbb{C}^{N_r}$ represents the radar beamformer,
- $\mathbf{R}_{t,x} = \mathbf{V}(\Delta f, R_t, \theta_t) \mathbf{X} \mathbf{X}^H \mathbf{V}^H(\Delta f, R_t, \theta_t)$,
- $\mathbf{R}_{n,x} = \sum_{q=1}^Q \gamma_q^2 \mathbf{V}(\Delta f, R_q, \theta_q) \mathbf{X} \mathbf{X}^H \mathbf{V}^H(\Delta f, R_q, \theta_q) + \sigma_{zr}^2 \mathbf{I}_{N_r N}$,
- $\mathbf{R}_{t,u} = \mathbf{V}(\Delta f, R_t, \theta_t)^H \mathbf{W} \mathbf{W}^H \mathbf{V}(\Delta f, R_t, \theta_t)$,
- $\mathbf{R}_{n,u} = \sum_{q=1}^Q \gamma_q^2 \mathbf{V}(\Delta f, R_q, \theta_q)^H \mathbf{W} \mathbf{W}^H \mathbf{V}(\Delta f, R_q, \theta_q) + \frac{\sigma_{zr}^2}{EN} \mathbf{W}^H \mathbf{W} \mathbf{I}_{N_t N}$, wherein E is the total transmit power.

C. Communication model

Assume a scenario involving U single-antenna communication users, all situated within the coverage of the FDA-MIMO DFRC transmitter. The received signal of the u -th user can be expressed as

$$y_u[n] = \mathbf{h}_u^T \mathbf{x}[n] + n_u[n], \quad (6)$$

where $n_u[n] \sim \mathcal{CN}(0, \sigma_u^2)$ represents the communication received noise, and $\mathbf{h}_u \in \mathbb{C}^{N_t}$ is the FDA-MIMO communication channel matrix, as

$$\mathbf{h}_u = \alpha_u \mathbf{A}(\Delta f, R_u, \theta_u) + \sqrt{\frac{1}{N_p}} \sum_{l=1}^{N_p} \alpha_{u,l} \mathbf{A}(\Delta f, R_{u,l}, \phi_{u,l}^t), \quad (7)$$

where N_p is the number of multi-path propagations, α_u and $\alpha_{u,l}$ are the channel coefficients of the LoS path and the l -th multipath, respectively, R_u and $R_{u,l}$ are the transmission range, θ_u and $\phi_{u,l}$ are the LoS angle and l -th angle-of-departure.

Assuming the expected symbol of the u -th user is $s_u[n]$, which requires a signal-to-noise ratio (SNR) of Ψ_u for transmission. Consequently, the information embedding can be implemented by interference-enhanced symbol-level precoding constraints as [10]

$$\begin{aligned} \text{C1: } \Re \left\{ \mathbf{h}_u^T \mathbf{x}[n] e^{-j\angle s_u[n]} - \sigma_u \sqrt{\Psi_u} \right\} \sin \Phi \\ - \left| \Im \left\{ \mathbf{h}_u^T \mathbf{x}[n] e^{-j\angle s_u[n]} \right\} \right| \cos \Phi \geq 0, \end{aligned} \quad (8)$$

where $\Phi = \frac{\pi}{M}$ is half of the beneficial interference angle region, M is the number of phase modulation constellations.

D. Problem formulation

To enhance the LPI performance, it is desirable to suppress the instantaneous transmit power toward the target across the pulse duration. Accordingly, the min-max criterion is adopted to minimize the peak transmitting beamforming level while jointly ensuring radar detection reliability and communication quality-of-service. We formulate the optimization problem of

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$$\begin{aligned} \min_{\mathbf{W}, \mathbf{X}} \quad & \max_n P_{\text{tran}}(R_t, \theta_t, \Delta f, \mathbf{x}[n]), \\ \text{s.t. :} \quad & \begin{cases} \text{C1 : } \Re \{ \mathbf{h}_u^T \mathbf{x}[n] e^{-j\angle s_u[n]} - \sigma_u \sqrt{\Psi_u} \} \sin \Phi \\ \quad - |\Im \{ \mathbf{h}_u^T \mathbf{x}[n] e^{-j\angle s_u[n]} \}| \cos \Phi \geq 0, \\ \text{C2 : } \text{SINR}(\mathbf{W}, \mathbf{X}) \geq \delta_t, \\ \text{C3 : } |\mathbf{X}(i)| = \sqrt{E/N_t}, \forall i, \\ \quad n = 1, \dots, N, u = 1, \dots, U, \end{cases} \end{aligned} \quad (9)$$

where δ_t is the radar SINR threshold.

III. JOINT WAVEFORM DESIGN AND BEAMPATTERN CONTROL ALGORITHM

A. Radar received beamformer

In this subsection, with the fixed transmitted waveform $\mathbf{X}$, solving the radar beamformer $\mathbf{W}$ only obeys the SINR constraint C2, which is equivalent to a MVDR beamforming problem, as [11]

$$\mathbf{W}^{(k)} = \frac{(\mathbf{R}_{nx})^{-1} \mathbf{V}(\Delta f, R_t, \theta_t) \mathbf{X}}{\mathbf{X}^H \mathbf{V}^H(\Delta f, R_t, \theta_t) (\mathbf{R}_{nx})^{-1} \mathbf{V}(\Delta f, R_t, \theta_t) \mathbf{X}}. \quad (10)$$

B. Transmit waveform

In this subsection, with the fixed radar receive beamformer $\mathbf{W}$, the transmit waveform is optimized, which reformulates the problem as

$$\begin{aligned} \min_{\mathbf{X}, \eta} \quad & \eta, \\ \text{s.t. :} \quad & \begin{cases} \text{C3 : } |\mathbf{X}(i)| = \sqrt{E/N_t}, \forall i, \\ \text{C4 : } \Re \{ \mathbf{H}_u^T \mathbf{X} - \mathbf{c}_u \} \sin \Phi - |\Im \{ \mathbf{H}_u^T \mathbf{X} \}| \cos \Phi \geq 0_N, \\ \text{C5 : } \mathbf{X}^H \mathbf{E}_n^T \tilde{\mathbf{A}}_{t,n} \mathbf{E}_n \mathbf{X} \leq \eta \\ \text{C6 : } \mathbf{X}^H (\delta_t \mathbf{R}_{nu} - \gamma_t^2 \mathbf{R}_{t,u}) \mathbf{X} \leq 0, \\ \quad n = 1, \dots, N, u = 1, \dots, U, \end{cases} \end{aligned} \quad (11)$$

where $\mathbf{c}_u = [\sigma_u \sqrt{\Psi_u}, \dots, \sigma_u \sqrt{\Psi_u}]^T \in \mathbb{C}^N$, and the expressions of $\mathbf{H}_u \in \mathbb{C}^{N_t N \times N}$, $\mathbf{E}_n \in \mathbb{C}^{N_t N \times N_t N}$, and $\tilde{\mathbf{A}}_{t,n} \in \mathbb{C}^{N_t N \times N_t N}$ are

$$\mathbf{H}_u = \text{blkdiag}(\mathbf{h}_u e^{-j\angle s_u[1]}, \dots, \mathbf{h}_u e^{-j\angle s_u[N]}), \quad (12)$$

$$\mathbf{E}_n = \text{blkdiag} \left(\mathbf{0}_{N_t}, \dots, \underbrace{\mathbf{I}_{N_t}}_{\text{the } n\text{-th block}}, \dots, \mathbf{0}_{N_t} \right), \quad (13)$$

$$\tilde{\mathbf{A}}_{t,n} = \text{blkdiag} \left(\mathbf{0}_{N_t}, \dots, \underbrace{\mathbf{A}_{t,n}}_{\text{the } n\text{-th block}}, \dots, \mathbf{0}_{N_t} \right). \quad (14)$$

To solve this problem, we propose a SDR-based two-step decomposition iterative algorithm, whose specific steps are given as follows:

Step I: Benchmark Waveform Construction:

To reformulate constant-modulus constraint C3 and quadratic constraints C5 and C6, the process of the SDR technique is to define the covariance matrix of $\mathbf{X}$ as $\mathbf{Z} = \mathbf{X} \mathbf{X}^H \in \mathbb{C}^{N_t N \times N_t N}$, which satisfies $\mathbf{Z} \succeq 0$. Beyond constraints C4, the SDR-based optimization problem of $\mathbf{Z}$ is

$$\begin{aligned} \min_{\mathbf{Z}, \eta} \quad & \eta, \\ \text{s.t. :} \quad & \begin{cases} \text{C7 : } \mathbf{Z} \succeq 0 \\ \text{C8 : } \mathbf{Z}(i, i) = E/N_t, \forall i, \\ \text{C9 : } \text{Tr}(\mathbf{E}_n^T \tilde{\mathbf{A}}_{t,n} \mathbf{E}_n \mathbf{Z}) \leq \eta \\ \text{C10 : } \text{Tr}((\delta_t \mathbf{R}_{nu} - \gamma_t^2 \mathbf{R}_{t,u}) \mathbf{Z}) \leq 0, \\ \quad n = 1, \dots, N, u = 1, \dots, U, \end{cases} \end{aligned} \quad (15)$$

Problem (15) is a convex semi-definite program (SDP) problem, which can be efficiently solved using interior-point algorithms [12]. After obtaining $\mathbf{Z}$, an initial waveform $\tilde{\mathbf{X}}$ is constructed to serve as a reference point for transmit waveform recovery. Specifically, we perform eigenvalue decomposition as

$$\mathbf{Z} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^H, \quad (16)$$

where $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_{N_t N}]$ satisfying $\mathbf{U}^H \mathbf{U} = \mathbf{I}_{N_t N}$ is a unitary matrix, and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_{N_t N})$ is the eigenvalue matrix with $\lambda_1 \geq \dots \geq \lambda_{N_t N}$.

It should be noted that $\tilde{\mathbf{X}} \tilde{\mathbf{X}}^H = \mathbf{Z}$ is required for rank($\mathbf{Z}$) = 1, and the optimal solution of problem (15) may not strictly guarantee this. To improve recovery accuracy, we define the characteristic density as

$$\varrho = \frac{\lambda_1}{\text{Tr}(\mathbf{\Lambda})}. \quad (17)$$

Then, the SDR solution can be derived into two case:

- $0.8 \leq \varrho \leq 1$: The SDR solution is strictly or nearly satisfying rank($\mathbf{Z}$) = 1. The initial waveform can be exactly recovered from the dominant eigenpair as $\tilde{\mathbf{X}} = \sqrt{\lambda_1} \mathbf{u}_1$.
- $\varrho < 0.8$: The SDR solution exhibits a pronounced high-rank feature. The Gaussian randomization is employed to generate candidate initial waveform sets, from which a feasible solution is selected. Specifically, construct a random Gaussian sets as

$$\tilde{\mathbf{X}}_\ell = \mathbf{U} \mathbf{\Lambda} \mathbf{G}_\ell, \ell = 1, \dots, \mathcal{L}, \quad (18)$$

where $\mathcal{L}$ is the number of randomization trials, and $\mathbf{G}_\ell \sim \mathcal{CN}(0, \mathbf{I}_{N_t N})$ is the Gaussian numerical matrix. For each projected candidate, the original constraints C7-C10 are evaluated. Among all feasible candidates, the one yielding the minimum objective value is selected as:

$$\tilde{\mathbf{X}} = \arg \min_{\tilde{\mathbf{X}}_\ell} \eta, \text{ s.t. : C7 - C10}. \quad (19)$$

Step II: Optimal Waveform Solution:

The feasible region of the transmitted waveform must satisfy the information embedding constraint C4, while the initial

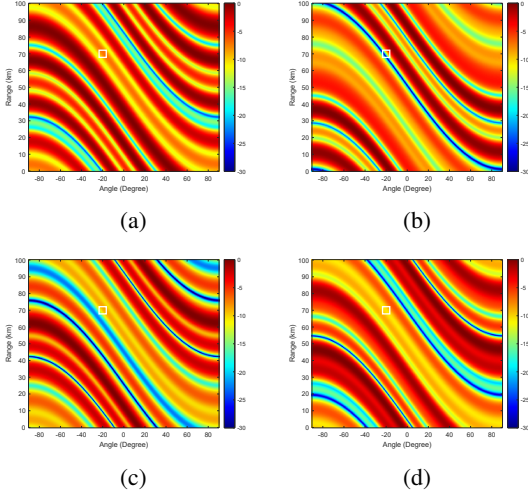


Fig. 1. Transmit beampattern. (a) $n = 1$, (b) $n = 12$, (c) $n = 22$, (d) $n = 32$.

waveform $\tilde{\mathbf{X}}$ obtained in Step I is used as a baseline to build an MSE problem, which is formulated as

$$\begin{aligned} \min_{\mathbf{X}} & \left\| \mathbf{X} - \tilde{\mathbf{X}} \right\|^2, \\ \text{s.t.} : & \begin{cases} \text{C4} : \Re \{ \mathbf{H}_u^T \mathbf{X} - \mathbf{c}_u \} \tan \Phi - |\Im \{ \mathbf{H}_u^T \mathbf{X} \}| \cos \Phi \geq \mathbf{0}_N, \\ n = 1, \dots, N, u = 1, \dots, U, \end{cases} \end{aligned} \quad (20)$$

Problem (20) is convex, which can be directly solved using interior-point algorithms. Finally, perform element-by-element constant modulus projection as

$$\mathbf{X}^{(k)}(i) = \sqrt{\frac{E}{N_t}} e^{j\angle \mathbf{X}^{(k)}(i)}. \quad (21)$$

IV. SIMULATION RESULTS

An FDA-MIMO DFRC system is considered, in which both the transmit and receive sides employ uniform linear arrays with $N_t = N_r = 10$ antennas and half-wavelength inter-element spacing. The carrier frequency is set to $f_0 = 1$ GHz, with a frequency increment of $\Delta f = 3$ kHz. The fast-time waveform is discretized into $N = 32$ samples over a pulse duration of $T_p = 5 \times 10^{-6}$ s. The target is located at (70 km, -20°), while two stationary clutter scatterers are located at (30 km, -50°) and (50 km, 40°), respectively. Moreover, three single-antenna communication users are positioned at (15 km, -50°), (44 km, -20°), and (80 km, 45°), and QPSK modulation is employed for symbol transmission.

Fig. 1 illustrates the range-angle dependent transmit beampattern across different fast-time slots, where the white rectangle indicates the target location. Due to the LPI-oriented design, the transmit power toward the target is intentionally suppressed over all time slots, resulting in a consistent power level below -10 dB. These results demonstrate that instantaneous power peaks are mitigated through the proposed

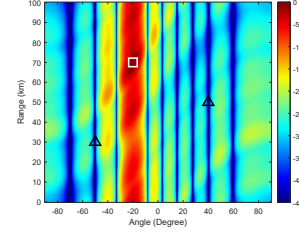


Fig. 2. Transceive beampattern.

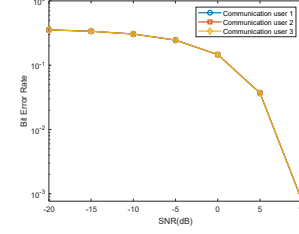


Fig. 3. Communication BERs of each user.

waveform design method, thereby reducing the interception probability while preserving waveform diversity.

Fig. 2 depicts the radar transceive beampattern, where the target and clutter locations are indicated by the white rectangle and black triangles, respectively. It is observed that the proposed algorithm achieves effective two-dimensional target focusing in range-angle while significantly suppressing clutter signals. Although the transmit power toward the target is deliberately reduced, effective target focusing is still achieved due to the coherent gain introduced by the receive beamforming. The proposed method allows the transmit power to be reduced without violating the radar SINR requirement, which enhances the LPI capability.

Fig. 3 presents the BER performance of the communication users, where each data point is obtained from 2000 Monte Carlo trials. It is observed that the BER of all users decreases rapidly as the signal-to-noise ratio increases, indicating fast convergence behavior. This observation demonstrates that the proposed symbol-level waveform design effectively exploits beneficial interference, thereby achieving stable and reliable multi-user communication.

V. CONCLUSION

This paper proposes a joint waveform design and transmit beampattern control framework for FDA-MIMO DFRC systems. The objective is to suppress the instantaneous power of the time-varying transmit beampattern while satisfying both radar detection and communication performance requirements. An MVDR-SDR-based alternative optimization algorithm is developed to update the radar received beam and integrated transmit waveform iteratively. Simulation results demonstrate that the proposed algorithm enables flexible time-varying transmitted beampattern control, effective clutter suppression, and stable communication BER performance.

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